

# LIBRARY BOOK REQUEST SYSTEM USING SERVICENOW

## Project Documentation

### Project Title

Library Book Request System Using ServiceNow

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## Abstract

The Library Book Request System is a ServiceNow-based application developed to automate the process of requesting, approving, and tracking library books within an educational institution. The system allows students to submit book requests electronically, while librarians can review and update request statuses. Automated email notifications keep users informed about the progress of their requests. The application improves efficiency, reduces manual paperwork, and provides transparency in library operations.

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## Objective

The primary objective of this project is to develop an automated platform that enables students to request books and allows librarians to manage approvals through a structured workflow in ServiceNow.

Objectives include:

- Automate book request management.
  - Reduce manual record maintenance.
  - Enable librarian approval workflow.
  - Send automated email notifications.
  - Maintain request history and status tracking.
  - Improve overall library service efficiency.
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# Milestone 1: Environment Setup

## Objective

To prepare the ServiceNow development environment and create the application foundation.

### Step 1: Create Personal Developer Instance

1. Create a ServiceNow Developer Account.
2. Request a Personal Developer Instance (PDI).
3. Access the instance using administrator credentials.

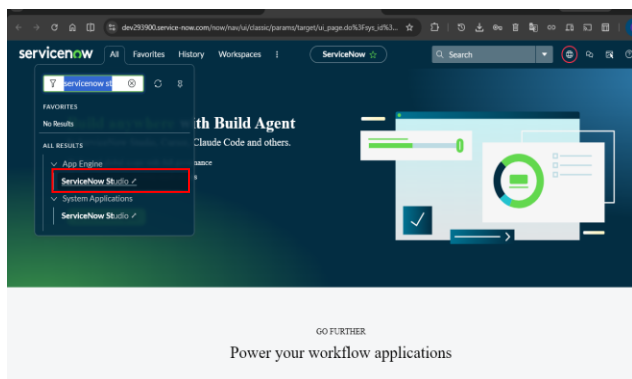
### Step 2: Create Scoped Application

Navigation:

System Applications → Studio

Application Name:

Library Book Request System



you can also add a thumbnail image.

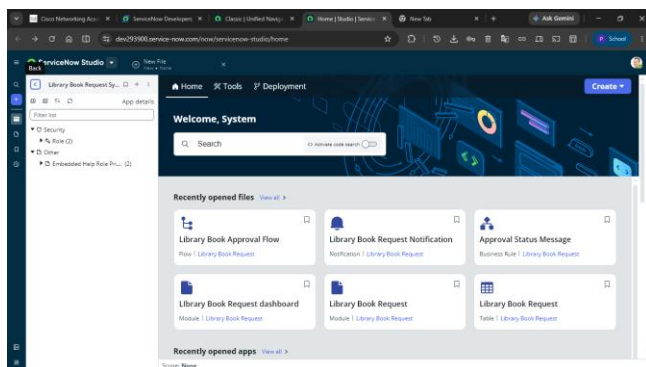
Name \*

Description

Scope \* ☒ Scoped ☐ Global

Browse or drag to upload

BMP, GIF, ICO, JPEG, JPG, PNG, SVG



## Learning Outcomes

- Understanding ServiceNow architecture.
- Creating a scoped application.
- Working with Studio.

# Milestone 2: Data Model Design

## Objective

Design custom tables required for library book management.

## Table Name

Library Book Request

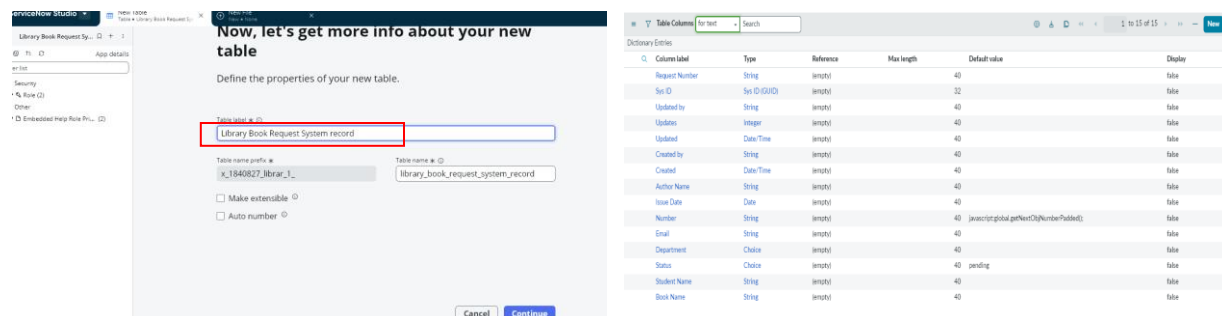
## Fields Created

| Field Name     | Type    |
|----------------|---------|
| Student Name   | String  |
| Request Number | Integer |
| Department     | Choice  |
| Book Name      | String  |
| Author Name    | String  |
| Issue Date     | Date    |
| Status         | Choice  |
| Email          | Email   |

## Status Values

- Pending
- Approved
- Rejected

## Screenshot



Create Choice List

Related Links

[Show Table](#)
[Run Point Scan](#)
[Advanced view](#)
[SN Utils | Versions \(0\)](#)

Access Controls

Choices (3)

Attributes

Labels (1)

≡

▼

Label

Search

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Actions on selected rows...

Choices

| <input type="checkbox"/> | Label     | Value     | Language | Sequence | Inactive | Updated             |
|--------------------------|-----------|-----------|----------|----------|----------|---------------------|
| <input type="checkbox"/> | Pending   | pending   | en       | 100      | false    | 2026-06-16 11:49:56 |
| <input type="checkbox"/> | Rejection | rejection | en       | 200      | false    | 2026-06-16 11:50:04 |
| <input type="checkbox"/> | Approved  | approved  | en       | 300      | false    | 2026-06-16 11:50:09 |

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## Learning Outcomes

- Created custom application table.
- Designed form fields.
- Implemented status tracking.

# Milestone 3: Form Configuration

## Objective

Configure request forms for user-friendly data entry.

## Form Layout Configuration

Navigation:

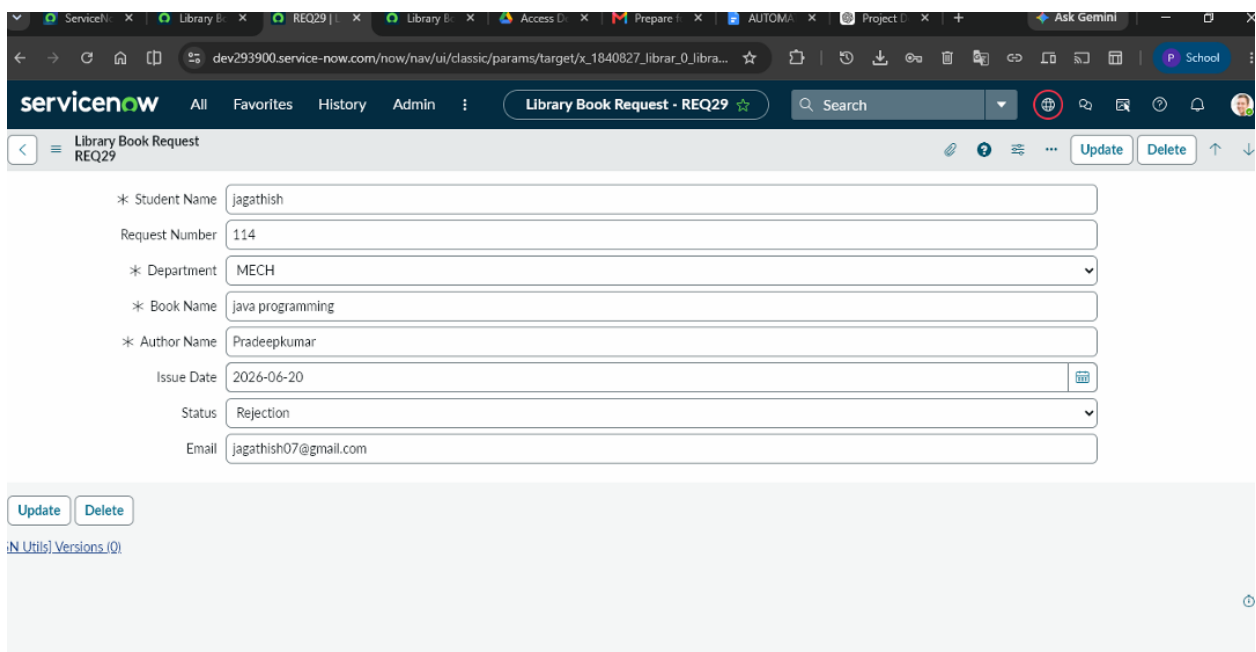
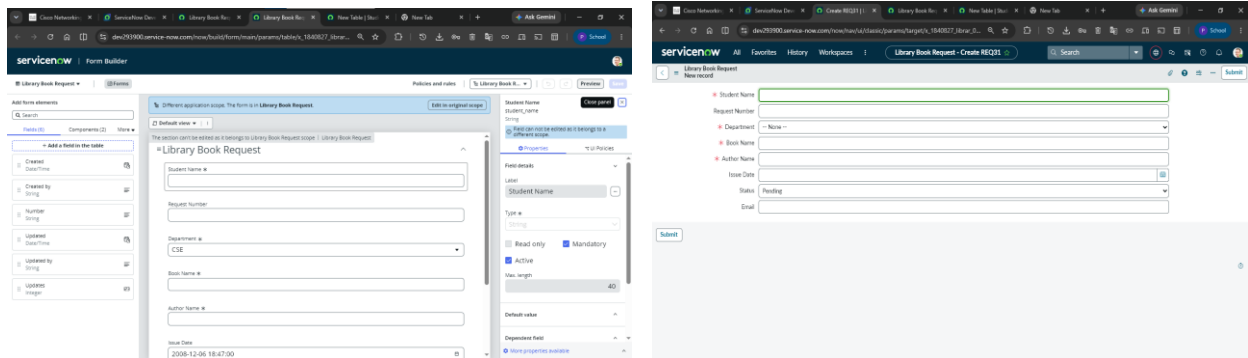
Open Library Book Request Form

Right Click → Configure → Form Layout

## Configured Fields

- Student Name
- Request Number
- Department
- Book Name
- Author Name
- Issue Date
- Status
- Email

## Screenshot



## Learning Outcomes

- Form customization.
- User-friendly interface design.

## Milestone 4: Flow Designer Implementation

### Objective

Automate approval processing through Flow Designer.

### Flow Name

Library Book Approval Flow

## Trigger

When a Library Book Request record is created or updated.

## Flow Logic

IF Status = Approved

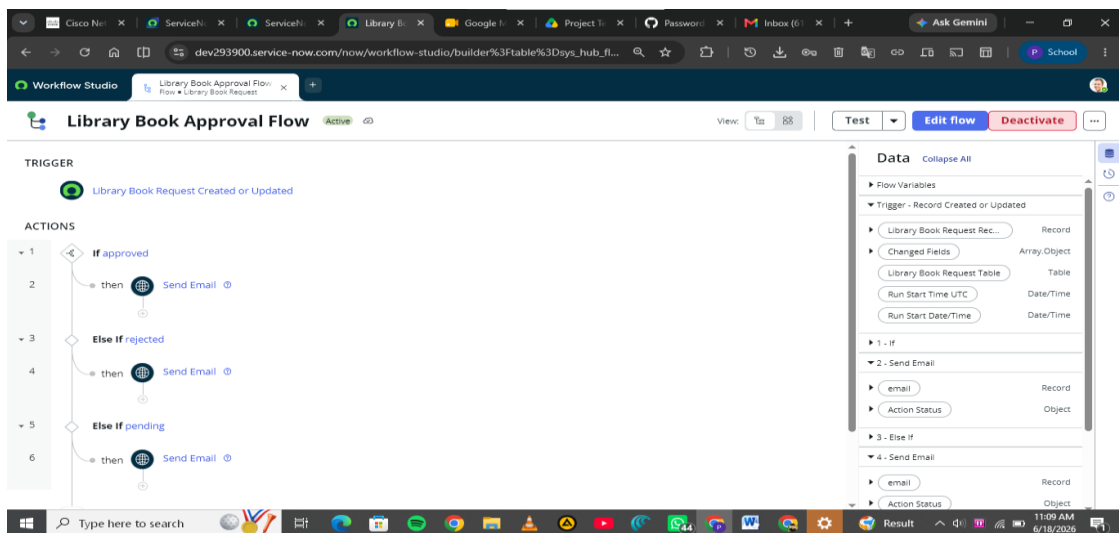
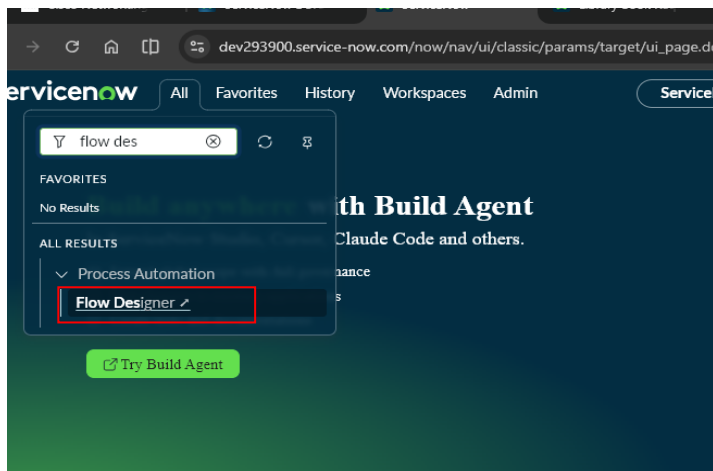
→ Send Approval Email

ELSE IF Status = Rejected

→ Send Rejection Email

ELSE IF Status = Pending

→ Send Submission Notification



## Learning Outcomes

- Created automated workflows.
  - Implemented decision-based processing.
  - Integrated notifications.
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## Milestone 5: Email Notifications

### Objective

Provide automated communication to students.

### Notification 1

Library Book Request Submitted Successfully

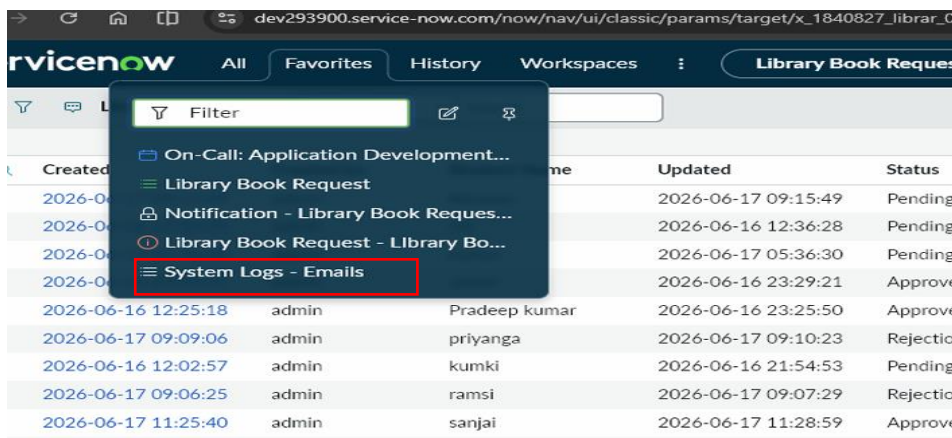
Recipient:

Student

Contents:

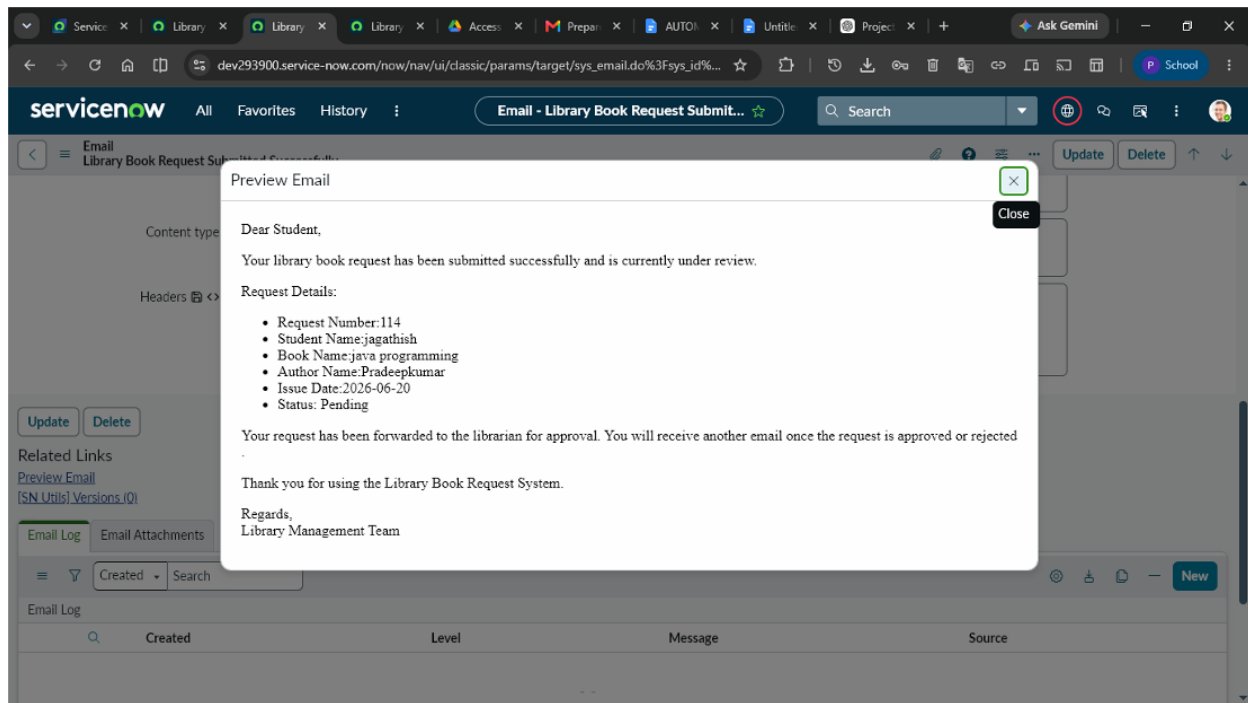
- Book Name
- Author Name
- Request Date
- Status : Pending and submitted

### Screenshot



The screenshot shows the ServiceNow interface for 'Library Book Request'. A dropdown menu is open, displaying several options. The option 'System Logs - Emails' is highlighted with a red box. The background shows a table with columns: Created, Name, Updated, and Status.

| Created             | Name  | Updated             | Status    |
|---------------------|-------|---------------------|-----------|
| 2026-06-16 12:25:18 | admin | 2026-06-16 23:25:50 | Approve   |
| 2026-06-17 09:09:06 | admin | 2026-06-17 09:10:23 | Rejection |
| 2026-06-16 12:02:57 | admin | 2026-06-16 21:54:53 | Pending   |
| 2026-06-17 09:06:25 | admin | 2026-06-17 09:07:29 | Rejection |
| 2026-06-17 11:25:40 | admin | 2026-06-17 11:28:59 | Approve   |



## Notification 2

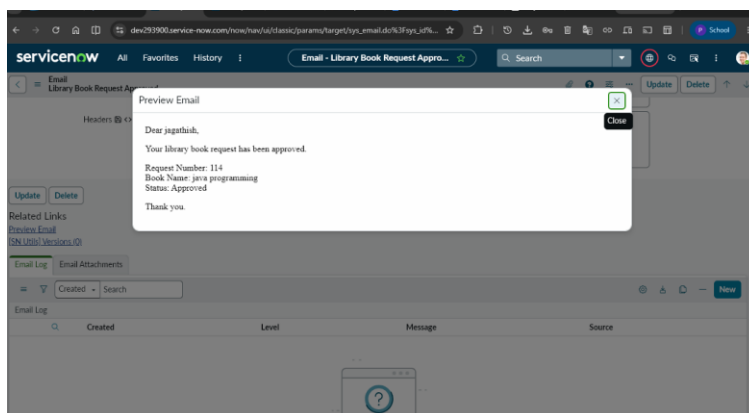
### Library Book Request Approved

Recipient:

Student

Contents:

- Request Number
- Approval Status
- Book Details
- Status : Approved





## Notification 3

### Library Book Request Rejected

Recipient:

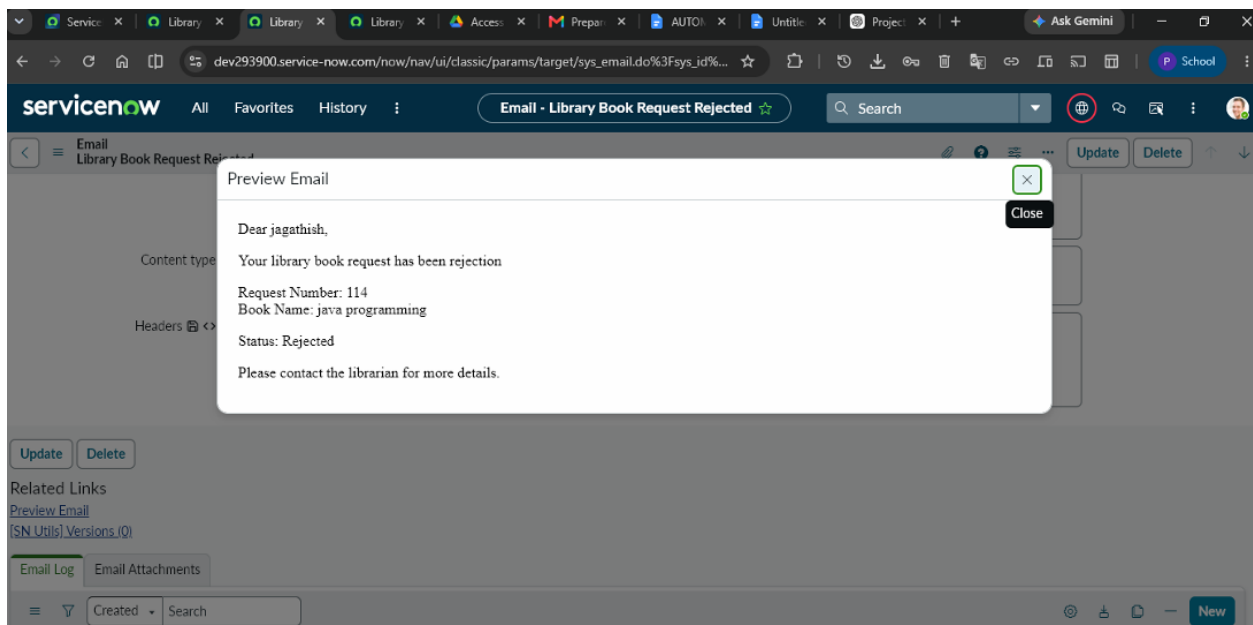
Student

Contents:

- Request Number
- Rejection Status

## Screenshot

### Email Notification Records



## Learning Outcomes

- Configured automated email notifications.
- Improved communication process.

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## Milestone 6: Request Management

### Objective

Provide librarians with a centralized view of requests.

### Features

- View all requests.

- Monitor request status.
- Update approval decisions.
- Track request history.

## Status Workflow

Pending → Approved

Pending → Rejected

| Created             | Created by | Student Name  | Updated             | Status    | Updated by | Updates | Department | Email                 |
|---------------------|------------|---------------|---------------------|-----------|------------|---------|------------|-----------------------|
| 2024-06-17 09:15:49 | admin      | Naveen        | 2024-06-17 09:15:49 | Pending   | admin      | 0       | CT         | naveen@gmail.com      |
| 2024-06-16 12:36:28 | admin      | riti          | 2024-06-16 12:36:28 | Pending   | admin      | 0       | IT         |                       |
| 2024-06-17 05:36:30 | admin      | kumar         | 2024-06-17 05:36:30 | Pending   | admin      | 0       | EEE        | 717823y174@kce.ac.in  |
| 2024-06-16 23:29:21 | admin      | vishal        | 2024-06-16 23:29:21 | Approved  | admin      | 0       | CYBER      |                       |
| 2024-06-16 12:25:18 | admin      | Pradeep kumar | 2024-06-16 23:25:50 | Approved  | admin      | 1       | ECE        |                       |
| 2024-06-17 09:09:06 | admin      | priyanga      | 2024-06-17 09:10:23 | Rejection | admin      | 1       | ADIS       | priyanga              |
| 2024-06-16 12:02:57 | admin      | kumki         | 2024-06-16 21:54:53 | Pending   | admin      | 1       | MBA        |                       |
| 2024-06-17 09:06:25 | admin      | ramsi         | 2024-06-17 09:07:29 | Rejection | admin      | 2       | IT         | ramsi@gmail.com       |
| 2024-06-17 11:25:40 | admin      | sanjal        | 2024-06-17 11:28:59 | Approved  | admin      | 2       | CIVIL      | sanjal@gmail.com      |
| 2024-06-17 11:50:07 | admin      | jagathish     | 2024-06-17 11:52:04 | Pending   | admin      | 2       | MECH       | jagathish07@gmail.com |
| 2024-06-16 12:28:22 | admin      | ram           | 2024-06-16 23:58:08 | Pending   | admin      | 2       | ECE        | 717823y134@kce.ac.in  |
| 2024-06-17 05:45:53 | admin      | viswarth      | 2024-06-17 06:27:42 | Pending   | admin      | 2       | CSE        | viswarth@gmail.com    |
| 2024-06-16 11:53:30 | admin      | kumar         | 2024-06-16 21:54:44 | Pending   | admin      | 2       | CD         |                       |
| 2024-06-17 05:09:34 | admin      | Nithish       | 2024-06-17 05:41:13 | Rejection | admin      | 3       | IT         | 717823y134@kce.ac.in  |
| 2024-06-17 09:20:41 | admin      | nithish       | 2024-06-17 09:22:30 | Pending   | admin      | 3       | CSE        | nithish@gmail.com     |

| Created             | Created by | Student Name  | Updated             | Status    | Updated by | Updates | Department | Email                 |
|---------------------|------------|---------------|---------------------|-----------|------------|---------|------------|-----------------------|
| 2024-06-17 09:15:49 | admin      | Naveen        | 2024-06-17 09:15:49 | Pending   | admin      | 0       | CT         | naveen@gmail.com      |
| 2024-06-16 12:36:28 | admin      | riti          | 2024-06-16 12:36:28 | Pending   | admin      | 0       | IT         |                       |
| 2024-06-17 05:36:30 | admin      | kumar         | 2024-06-17 05:36:30 | Pending   | admin      | 0       | EEE        | 717823y174@kce.ac.in  |
| 2024-06-16 23:29:21 | admin      | vishal        | 2024-06-16 23:29:21 | Approved  | admin      | 0       | CYBER      |                       |
| 2024-06-16 12:25:18 | admin      | Pradeep kumar | 2024-06-16 23:25:50 | Approved  | admin      | 1       | ECE        |                       |
| 2024-06-17 09:09:06 | admin      | priyanga      | 2024-06-17 09:10:23 | Rejection | admin      | 1       | ADIS       | priyanga              |
| 2024-06-16 12:02:57 | admin      | kumki         | 2024-06-16 21:54:53 | Pending   | admin      | 1       | MBA        |                       |
| 2024-06-17 09:06:25 | admin      | ramsi         | 2024-06-17 09:07:29 | Rejection | admin      | 2       | IT         | ramsi@gmail.com       |
| 2024-06-17 11:25:40 | admin      | sanjal        | 2024-06-17 11:28:59 | Approved  | admin      | 2       | CIVIL      | sanjal@gmail.com      |
| 2024-06-17 11:50:07 | admin      | jagathish     | 2024-06-17 11:52:04 | Rejection | admin      | 2       | MECH       | jagathish07@gmail.com |
| 2024-06-16 12:28:22 | admin      | ram           | 2024-06-16 23:58:08 | Pending   | admin      | 2       | ECE        | 717823y134@kce.ac.in  |
| 2024-06-17 05:45:53 | admin      | viswarth      | 2024-06-17 06:27:42 | Rejection | admin      | 2       | CSE        | viswarth@gmail.com    |
| 2024-06-16 11:53:30 | admin      | kumar         | 2024-06-16 21:54:44 | Approved  | admin      | 2       | CD         |                       |
| 2024-06-17 05:09:34 | admin      | Nithish       | 2024-06-17 05:41:13 | Rejection | admin      | 3       | IT         | 717823y134@kce.ac.in  |
| 2024-06-17 09:20:41 | admin      | nithish       | 2024-06-17 09:22:30 | Pending   | admin      | 3       | CSE        | nithish@gmail.com     |

## Learning Outcomes

- Request monitoring.
  - Approval management.
  - Record tracking.
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## Testing

### Test Case 1

Student submits a book request.

Expected Result:

Status becomes Pending.

Result:

Passed.

### Test Case 2

Librarian approves request.

Expected Result:

Status changes to Approved.

Result:

Passed.

### Test Case 3

Librarian rejects request.

Expected Result:

Status changes to Rejected.

Result:

Passed.

### Test Case 4

Email notification generated.

Expected Result:

Email sent successfully.

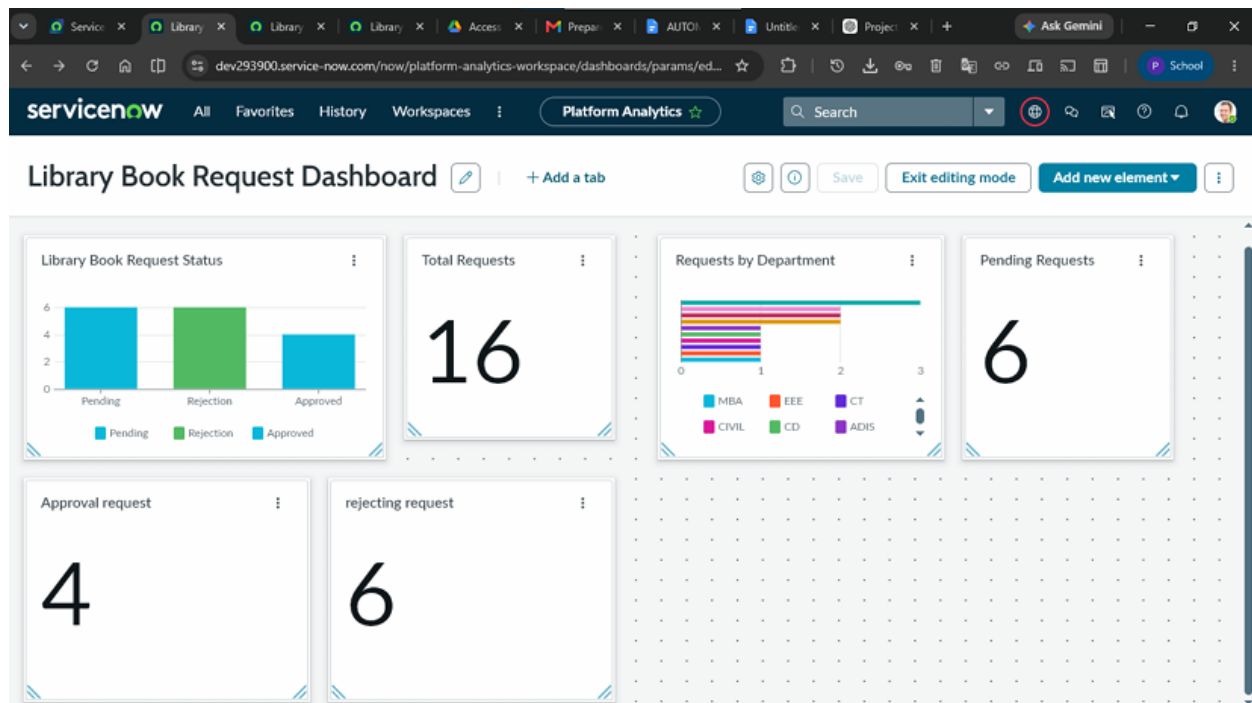
Result:

Passed.

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## Dashboard Components

- Total Requests
- Pending Requests
- Approval Requests
- Rejecting Requests
- Requests by Status
- Requests by Department



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## Results

The Library Book Request System successfully automated the book request lifecycle.

#### Achievements:

- Online request submission.
  - Automated approval workflow.
  - Real-time status tracking.
  - Email notifications.
  - Centralized request management.
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## Conclusion

The Library Book Request System developed using ServiceNow successfully automates the process of requesting and approving library books. The application eliminates manual paperwork, improves communication through automated notifications, and provides efficient management of library requests. The project demonstrates practical implementation of ServiceNow concepts including scoped applications, custom tables, Flow Designer, notifications, and request management.

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## Future Enhancements

- Multi-level librarian approval.
- Service Portal integration.
- Mobile application support.
- Book inventory management.
- Due date tracking.
- Fine management system.
- Dashboard analytics.
- AI-powered book recommendations.
- Integration with external library databases.